Caileigh Shoot

Munich, Germany
caileighshoot@gmail.com
linkedin.com/in/caileighshoot
Dual US/German Citizen

EDUCATION

University of Washington, M.S. Remote Sensing and Geospatial Analysis, 2016-2018
- *Thesis:* Classifying FIA Forest Type from a Fusion of Hyperspectral and LiDAR data

University of Washington, B.S., Environmental Science and Resource Management, 2011-2015

TECHNICAL SKILLS

- **UAVs and Remote Sensing:** Photogrammetry, Multispectral & Hyperspectral Imaging, LiDAR, UAV Piloting
- **Programming:** R (Advanced), Python, SQL, Command Line, Git
- **Machine Learning and Statistics:** Bayesian, K Nearest Neighbor, Maximum Likelihood, Random Forest, SVM
- **Software:** Agisoft Photoscan, ArcGIS, Drone Deploy, ESRI Field Maps, FUSION (LiDAR), Global Mapper LiDAR Module, Google Earth Pro, Google Earth Engine, Pix4D, QGIS, R Studio, Survey 123

EXPERIENCE

Technische Universität Berlin, Research Associate (KI-Recover) — August 2025 - Present
- Conduct research on forest recovery in Germany following large-scale disturbances including bark beetle infestations, windthrow, and fire
- Apply remote sensing, GIS, and ecological analysis to assess forest resilience and regeneration patterns

Weyerhaeuser, Lead Forest Carbon Scientist — December 2022 – July 2025
- Lead the design and implementation of high-integrity forest carbon projects under voluntary carbon market standards (e.g., Verra, ACR), ensuring methodological compliance and scientific rigor
- Apply advanced data science techniques to analyze forest inventory, remote sensing (e.g., LiDAR, Sentinel-2), and GIS data to inform carbon stock assessments and growth and yield modeling
- Create custom remote sensing and GIS-based decision support tools using R Shiny, Power BI, ArcGIS Web Dashboards, and more; One tool delivered >$1.3M in cost savings and received Weyerhaeuser's 2025 Innovator of the Year award
- Develop and standardize internal protocols, including inventory sampling design, project development plans, and technical documentation rooted in peer-reviewed literature and evolving methodologies
- Assist with field data collection and on-site visits with carbon project auditors

University of Washington, Remote Sensing Lecturer — September 2022 – December 2022
- Lead a class of 115 students in weekly lectures for the Remote Sensing of the Environment (ESRM 430) course, a required class for Environmental Science students at the University of Washington
- Provide students with guidance and mentorship, providing insights into how they can use remote sensing in their future careers and helping to apply every lesson to their future

Weyerhaeuser, Inventory and Planning GIS Analyst — May 2021 – December 2022
- Support fast-paced forest operations with GIS mapping and spatial analysis
- Use SQL and Python to develop tools and analyze large forest inventory data sets
- Provide user support and training for internal GIS applications and databases

Weyerhaeuser, Remote Sensing Research Specialist — September 2018 – November 2020, March - May 2021
- Develop and implement forestry research studies that integrate innovative remote sensing technology and analysis and provided strong business value to Weyerhaeuser, including multiple high-level proprietary research projects
- Collect, process, and analyze multispectral UAV imagery and provide technical support and training for other UAV users across the company, while also maintaining multiple industrial-grade UAVs and sensors

NASA Goddard Space Flight Center, Arctic Boreal Vulnerability Experiment Airborne Campaign Data Fusion Intern — September 2018 – November 2020

- Use hyperspectral and LiDAR data to identify and evaluate aspen leaf miner damage in interior Alaska
- Process high resolution remote sensing data through NASA's supercomputer via R scripts and SLURM

UW Remote Sensing and Geospatial Analysis Laboratory, Laboratory Manager and Graduate Research Assistant — September 2014 – June 2018

- Analyze hyperspectral and LiDAR data with geospatial and statistical analysis software and build machine-learning based (i.e., Random Forest, SVM, KNN) forest type prediction models
- Assist with various research projects in the Remote Sensing and Geospatial Analysis Lab, including the collection of field spectrometer readings of a variety of vegetation in interior Alaska
- Served as teaching assistant for Remote Sensing of the Environment (ESRM 430) from January – March 2016, leading 4 weekly technical lab sessions, grading student papers, and holding regular office hours
- Serve as field crew lead from May to July 2015, obtaining a variety of ecological (stream temperature, solar radiation, etc.) and geospatial measurements (GPS Locations, Total Station maps, etc.) with the highest possible precision

MicaSense, GIS Analyst — September 2015 – June 2016

- Process multispectral imagery in an internal cloud-based platform and provide customer support for data quality issues in a fast-paced start up environment

UW Remote Sensing and Geospatial Analysis Laboratory, Field Crew Lead — May 2015 – July 2015
PI: Professor L. Monika Moskal

- Hire crew, organized gear, and manage the acquisition, storage, and backup of all data
- Obtain a variety of ecological (stream temperature, solar radiation, etc.) and geospatial measurements (GPS Locations, Total Station maps, etc.) with the highest possible precision
- Work resulted in the 2019 publication "Lidar-Based Approaches for Estimating Solar Insolation in Heavily Forested Streams"

Ecosystem Structure and Function Laboratory, Field Research Assistant — July – September 2014

- Collect data for forestry research projects overseen by Dr. Derrick Churchill, using a variety of field equipment such as a Total Station, Laser Range Finder, Diameter Tape, and more
- Work resulted in the 2018 publication "Applying LiDAR Individual Tree Detection to Management of Structurally Diverse Forest Landscapes"

UW Sustainability, Green Laboratory Program Coordinator — May 2012 – June 2015

- Develop program dedicated to improving sustainability of campus laboratories
- Manage multiple teams of interns, maintain web page, and coordinate outreach
- Serve on multiple committees dedicated to improving sustainability metrics

PUBLICATIONS

Montenegro, J.F., Mohan, M., Ewane, E.B., Friess, D.A., Selvam, P.P., Roy, A.D., Althauser, A., Tidwell, J., Gamboa-Cutz, J., **Shoot, C.** & Watt, M.S., (2025). Mangrove-based carbon market projects: Current trends and future perspectives. *Forest Policy and Economics*, *181*, 103658.

Shoot, C., Andersen, H. E., Moskal, L. M., Babcock, C., Cook, B. D., & Morton, D. C. (2021). Classifying forest type in the national forest inventory context with airborne hyperspectral and lidar data. *Remote Sensing*, 13(10), 1863.

CONTRIBUTIONS

Holub, S. M., Catnach, G., Littke, K. M., & Hatten, J. A. (2024). Forest soil carbon storage in 10-year-old Douglas-fir plantations of western Oregon and Washington remains similar to pre-harvest. *Soil Science Society of America Journal*.

Richardson, J. J., Torgersen, C. E., & Moskal, L. M. (2019). Lidar-based approaches for estimating solar insolation in heavily forested streams. *Hydrology and Earth System Sciences*, *23*(7), 2813-2822.

Jeronimo, S. M., Kane, V. R., Churchill, D. J., McGaughey, R. J., & Franklin, J. F. (2018). Applying LiDAR individual tree detection to management of structurally diverse forest landscapes. *Journal of Forestry*, *116*(4), 336-346.

PRESENTATIONS

Applications for Remote Sensing in the Voluntary Forest Carbon Market
- ForestSAT 2024 | Rotorua, New Zealand | September 10, 2024

The Science and Art of LiDAR
- Maptime Seattle | Seattle, WA | October 4, 2023

A Practical Overview of Geospatial R
- Maptime Seattle | Seattle, WA | April 4, 2018
- 2018 Washington GIS Conference | Olympia, WA | May 22, 2018

Working With Lidar
- 2018 Washington GIS Conference | Olympia, WA | May 22, 2018

Classifying FIA Forest Type from Hyperspectral and Lidar Data
- 2017 Forest Inventory and Analysis Stakeholder Science Meeting | Park City, UT, USA |October 24-26, 2017

Airborne Lidar Accurately Detects Low-Stature Understory Vegetation
- Symposium on Systems Analysis in Forest Resources | Portland, OR, USA | August 6-11, 2017
- Ecological Society of America 2017 Annual Meeting | Suquamish, WA, USA | August 27-30, 2017
- ForestSAT 2016 | Santiago, Chile | November 14-18, 2016
- EARSeL 3rd Workshop SIG on Forestry | Krakow, Poland | September 12-16, 2016
- American Society for Photogrammetry and Remote Sensing Puget Sound Region 2015 Spring Info Session and Tech Exchange | Burnaby, British Columbia, Canada | April 8th, 2015

A Simple GIS Workflow in Python and R
- CUGOS Spring Fling | Seattle, WA, USA | May 20, 2017

AWARDS

- Innovator of the Year, Weyerhaeuser, 2025
- SEFS Graduate Research Fellowship, University of Washington, 2016-2017
- ESRM Senior Capstone Award, University of Washington, 2015